

OptiCrop - Smart Agricultural Production Optimization Engine

Purpose

The OptiCrop project optimizes agricultural production using machine learning and data-driven analysis of Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall. It recommends suitable crops, evaluates crop suitability, and supports researchers and policymakers with sustainable farming decisions.

Features

- Smart crop recommendation.
- Crop suitability and environmental assessment.
- Maximum yield prediction.
- Research and policy support.
- Sustainable farming insights.
- User-friendly web interface.
- Fast predictions using Pickle (.pkl).

Scenario 1: Farmers enter soil/environment data; system recommends the best crop.

Scenario 2: Users assess whether conditions suit a particular crop.

Scenario 3: Researchers analyze crop-environment relationships for policy planning.